

Affine and Linear equivalence algorithms

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Abstract

The abstract serves both as a general introduction to the topic and as a brief, non-technical summary of the main results and their implications. Authors are advised to check the author instructions for the journal they are submitting to for word limits and if structural elements like subheadings, citations, or equations are permitted.

Keywords: keyword1, Keyword2, Keyword3, Keyword4

1 Equivalence

$$G = A_2 \circ F \circ A_1$$

Given functions $\mathbb{F}_q^n \rightarrow \mathbb{F}_q^m$.

2 Ordered partition

An *ordered partition* of a set I is an ordered tuple $\Pi = (B_1, \dots, B_k)$ of subsets of I where $\bigcup B_i = I$.

Two ordered partitions $\Pi = (B_1, \dots, B_k)$ and $\Pi' = (B'_1, \dots, B'_{k'})$ have the same *shape* if $k = k'$ and $|B_j| = |B'_j|$ for all $j \in \{1, \dots, k\}$.

We say that $\Pi' = (B'_1, \dots, B'_{k'})$ is a *refinement* of $\Pi = (B_1, \dots, B_k)$ if Π' is obtained by splitting some of the blocks of Π into smaller pieces, without changing their order. Concretely, each block B_i is a disjoint union of some consecutive blocks of Π' .

3 The search state

The algorithm will maintain two partial affine maps $A_1 : D_1 \subseteq \mathbb{F}_q^n \rightarrow \mathbb{F}_q^n$ and $A_2 : D_2 \subseteq \mathbb{F}_q^m \rightarrow \mathbb{F}_q^m$.

We keep ordered partitions for each function. For F we maintain an *input-point* partition $\mathcal{P}_F = (P_1^F, \dots, P_k^F)$, an *output-point* partition $\mathcal{Q}_F = (Q_1^F, \dots, Q_{k'}^F)$, and the hyperplane partitions $\mathcal{L}_F = (L_1^F, \dots, L_l^F)$ and $\mathcal{R}_F = (R_1^F, \dots, R_r^F)$. For G we maintain the analogous quadruple $\mathcal{P}_G = (P_1^G, \dots, P_k^G)$, $\mathcal{Q}_G = (Q_1^G, \dots, Q_{k'}^G)$, $\mathcal{L}_G = (L_1^G, \dots, L_l^G)$, and $\mathcal{R}_G = (R_1^G, \dots, R_r^G)$. Here $\mathcal{P}_F, \mathcal{P}_G$ are partitions of $\mathbb{F}_q^n$, $\mathcal{Q}_F, \mathcal{Q}_G$ are partitions of $\mathbb{F}_q^m$, $\mathcal{L}_F, \mathcal{L}_G$ are partitions of $AGr(n-1, n)$, and $\mathcal{R}_F, \mathcal{R}_G$ are partitions of $AGr(m-1, m)$. The algorithm will maintain the following invariants for these ordered partitions; given the current partial maps (A_1, A_2) , any extension $(\widehat{A}_1, \widehat{A}_2)$ satisfies

Domain hyperplane invariant. $\widehat{A}_1$ respect the domain hyperplane partitions $\mathcal{L}_F$ and $\mathcal{L}_G$

$$\widehat{A}_1(L_i^F) = L_i^G.$$

Codomain hyperplane invariant. $\widehat{A}_2$ respects the the codomain-hyperplane partitions $\mathcal{R}_F$ and $\mathcal{R}_G$: if $R_i^F \in \mathcal{R}_F$ is the i -th cell, then

$$\widehat{A}_2(R_i^F) = R_i^G.$$

Input point-cell invariant. $\widehat{A}_1$ respects the input-point partitions $\mathcal{P}_F$ and $\mathcal{P}_G$: if $x \in D_1 \cap P_i^F$, then $\widehat{A}_1(x) \in P_i^G$.

Output point-cell invariant. $\widehat{A}_2$ respects the output-point partitions $\mathcal{Q}_F$ and $\mathcal{Q}_G$: if $y \in D_2 \cap Q_i^F$, then $\widehat{A}_2(y) \in Q_i^G$.

Finally the current state will also maintain the current stabilizer of the the partial map (A_1, A_2) , of the automorphism group of G .

4 Signatures

Let $\mathcal{X}$ be a set of objects a *signature* is a map

$$sig : \mathcal{X} \rightarrow \Sigma$$

where Σ is an ordered set.

Definition 1 (Signature refinement of an ordered partition) Let $\Pi = (B_1, \dots, B_k)$ be an ordered partition of a set $\mathcal{X}$, and let $sig : \mathcal{X} \rightarrow \Sigma$ be a signature into an ordered set $\Sigma = (s_1, \dots, s_m)$. For each block B_i and each $s \in \Sigma$, set

$$B_{i,s} := \{x \in B_i : sig(x) = s\}.$$

The *refinement* of Π by sig is the ordered partition

$$\text{Ref}_{sig}(\Pi) := (B_{1,s_1}, \dots, B_{1,s_m}, B_{2,s_1}, \dots, B_{2,s_m}, \dots, B_{k,s_1}, \dots, B_{k,s_m}),$$

with empty blocks omitted.

For two signatures $sig_F, sig_G : \mathcal{X}_{\mathcal{F}} \rightarrow \Sigma$ with a relation $\sim$ on $\mathcal{X} \times \mathcal{X}$ we say that (sig_F, sig_G) is a *valid signature pair* if $x \sim y \implies sig_F(x) = sig_G(y)$.

In our case the objects $\mathcal{X}_F$ are either points or hyperplanes, and we will define the relation relative to a partial map $A = (A_1, A_2)$. We write $\sim_A$ for this relation.

- For points p, p' we say that $p \sim_A p'$ if there exist an extension $\hat{A} = (\hat{A}_1, \hat{A}_2)$ of A such that $\hat{A}_1(p) = p'$.
- For points q, q' in $\mathbb{F}_q^m$ we say that $q \sim_A q'$ if there exist an extension $\hat{A}$ of A such that $\hat{A}_2(q) = q'$.
- For hyperplanes U, U' in $\mathbb{F}_q^n$ we say that $U \sim_A U'$ if there exist an extension $\hat{A}$ of A such that $\hat{A}_1(U) = U'$.
- For hyperplanes V, V' in $\mathbb{F}_q^m$ we say that $V \sim_A V'$ if there exist an extension $\hat{A}$ of A such that $\hat{A}_2(V) = V'$.

We say that (sig_F, sig_G) is a *valid point/domain hyperplane/co-domain hyperplane signature pair* if its a valid signature given the relation $\sim_A$ for points, domain hyperplanes and co-domain hyperplanes respectively.

. if for the partial map $A = (A_1, A_2)$ we have to points $x \not\sim_A y$ then no extension map x to y so we can safely put x and y in non corresponding cells of $\mathcal{P}_F$ and $\mathcal{P}_G$ respectively without breaking the [Input point-cell invariant](#). The same logic also applies to the hyperplane partitions. The following Lemma describes this.

Lemma 1 (Refinement preserves alignment) Let $A = (A_1, A_2)$ be a partial map and (sig_F, sig_G) a valid signature pair (with respect to $\sim_A$). If an aligned pair of ordered partitions is refined using sig_F and sig_G , then the corresponding alignment invariant is preserved. More precisely:

1. (Input points) If $(\mathcal{P}_F, \mathcal{P}_G)$ satisfies [Input point-cell invariant](#), then $(\text{Ref}_{sig_F}(\mathcal{P}_F), \text{Ref}_{sig_G}(\mathcal{P}_G))$ also satisfies [Input point-cell invariant](#).
2. (Output points) If $(\mathcal{Q}_F, \mathcal{Q}_G)$ satisfies [Output point-cell invariant](#), then $(\text{Ref}_{sig_F}(\mathcal{Q}_F), \text{Ref}_{sig_G}(\mathcal{Q}_G))$ also satisfies [Output point-cell invariant](#).
3. (Domain hyperplanes) If $(\mathcal{L}_F, \mathcal{L}_G)$ satisfies [Domain hyperplane invariant](#), then $(\text{Ref}_{sig_F}(\mathcal{L}_F), \text{Ref}_{sig_G}(\mathcal{L}_G))$ also satisfies [Domain hyperplane invariant](#).
4. (Codomain hyperplanes) If $(\mathcal{R}_F, \mathcal{R}_G)$ satisfies [Codomain hyperplane invariant](#), then $(\text{Ref}_{sig_F}(\mathcal{R}_F), \text{Ref}_{sig_G}(\mathcal{R}_G))$ also satisfies [Codomain hyperplane invariant](#).

Proof We treat the input-point case; the other cases follow verbatim with the appropriate substitutions.

Input points. Let $A = (A_1, A_2)$ be a partial map, and let $\mathcal{P}_F = (P_1^F, \dots, P_k^F)$ and $\mathcal{P}_G = (P_1^G, \dots, P_k^G)$ be ordered input-point partitions satisfying [Input point-cell invariant](#). Let (sig_F, sig_G) be a valid signature pair for input points, and refine to obtain $\mathcal{P}'_F = \text{Ref}_{sig_F}(\mathcal{P}_F)$ and $\mathcal{P}'_G = \text{Ref}_{sig_G}(\mathcal{P}_G)$.

Let $\hat{A} = (\hat{A}_1, \hat{A}_2)$ be any extension of A , and suppose $\hat{A}_1(x) = y$ for some input point x . Then $x \sim_A y$, so validity gives $sig_F(x) = sig_G(y) = s$. If $x \in P_i^F$ then [Input point-cell invariant](#) implies $y \in P_i^G$, hence $x \in P_{i,s}^F$ and $y \in P_{i,s}^G$ in the refined partitions. Therefore [Input point-cell invariant](#) is preserved by refinement. $\square$

We are now going to go into details on the signatures of will use. The only thing we need to prove now is that the signatures are valid, and the correctness of our algorithms will follow.

4.1 Valid signatures

Lemma 2 (Composition preserves validity) If (sig_F, sig_G) is a valid signature pair and π is any function on the codomain of the signatures, then $(\pi \circ sig_F, \pi \circ sig_G)$ is also a valid signature pair.

Proof Let (A_1, A_2) be a partial map and $(\widehat{A}_1, \widehat{A}_2)$ be any extension. If x is mapped to x' (in the appropriate domain), validity of (sig_F, sig_G) gives $sig_F(x) = sig_G(x')$. Applying π to both sides yields $(\pi \circ sig_F)(x) = (\pi \circ sig_G)(x')$, as required. $\square$

Lemma 3 (Products preserve validity) If $(sig_F^{(1)}, sig_G^{(1)})$ and $(sig_F^{(2)}, sig_G^{(2)})$ are valid signature pairs (for the same object type), then the pair

$$(\langle sig_F^{(1)}, sig_F^{(2)} \rangle, \langle sig_G^{(1)}, sig_G^{(2)} \rangle)$$

is also a valid signature pair, where $\langle s_1, s_2 \rangle(x) := (s_1(x), s_2(x))$.

Proof Let (A_1, A_2) be a partial map and $(\widehat{A}_1, \widehat{A}_2)$ any extension. If x is mapped to x' , then validity of the two pairs gives $sig_F^{(1)}(x) = sig_G^{(1)}(x')$ and $sig_F^{(2)}(x) = sig_G^{(2)}(x')$. Therefore $\langle sig_F^{(1)}, sig_F^{(2)} \rangle(x) = \langle sig_G^{(1)}, sig_G^{(2)} \rangle(x')$, so the product pair is valid. $\square$

Finally we need a transfer the following transfer Lemma.

Lemma 4 For a partial map $A = (\widehat{A}_1, \widehat{A}_2)$ let $\widehat{A} = (\widehat{A}_1, \widehat{A}_2)$ be any extension of A then

1. For every $p \in \mathbb{F}_q^n$ and $U \in V_q(n)$

$$p \in U \Leftrightarrow \widehat{A}_1(p) \in \widehat{A}_1(U)$$

2. For every $p \in \mathbb{F}_q^n$ and $V \in V_q(m)$

$$p \in F^{-1}(V) \Leftrightarrow \widehat{A}_1(p) \in G^{-1}(\widehat{A}_2(V))$$

3. For every $U \in V_q(n)$ and $V \in V_q(m)$

$$|U \cap F^{-1}(V)| = |\widehat{A}_1(U) \cap G^{-1}(\widehat{A}_2(V))|$$

For each function, we maintain four ordered partitions:

1. $\mathcal{P}$: domain points,
2. $\mathcal{Q}$: co-domain points,
3. $\mathcal{L}$: domain flats,
4. $\mathcal{R}$: co-domain flats.

and each of them gets their own signatures. Thought this section fix the ordered partitions like above.

4.2 Point signatures

Let $p \in \mathbb{F}_q^n$ be any point, then we define the following signature:

$$\text{sig}_{\mathcal{L}}(p) = (|\{U \in L_1 : p \in U\}|, \dots, |\{U \in L_l : p \in U\}|)$$

while for a point $q \in \mathbb{F}_q^m$ we define the signature

$$\text{sig}_{\mathcal{R}}(q) = (|\{V \in R_1 : q \in V\}|, \dots, |\{V \in R_r : q \in V\}|)$$

Lemma 5 (Point signature validity)

- (i) Assume $(\mathcal{L}_F, \mathcal{L}_G)$ satisfies **Domain hyperplane invariant**. Then $(\text{sig}_{\mathcal{L}_F}, \text{sig}_{\mathcal{L}_G})$ is a valid input-point signature pair.
- (ii) Assume $(\mathcal{R}_F, \mathcal{R}_G)$ satisfies **Codomain hyperplane invariant**. Then $(\text{sig}_{\mathcal{R}_F}, \text{sig}_{\mathcal{R}_G})$ is a valid output-point signature pair.

Proof (i). Let $(\widehat{A}_1, \widehat{A}_2)$ be any extension. For $p \in \mathbb{F}_q^n$ write $p' := \widehat{A}_1(p)$. Fix i . By domain-hyperplane alignment, $U \mapsto \widehat{A}_1(U)$ is a bijection $L_i^F \rightarrow L_i^G$, and by Lemma 4(1) we have $p \in U \iff p' \in \widehat{A}_1(U)$. Therefore

$$|\{U \in L_i^F : p \in U\}| = |\{U' \in L_i^G : p' \in U'\}|.$$

Since this holds for all i , $\text{sig}_{\mathcal{L}_F}(p) = \text{sig}_{\mathcal{L}_G}(p')$.

(ii). For $q \in \mathbb{F}_q^m$ write $q' := \widehat{A}_2(q)$. Fix j . By codomain-hyperplane alignment, $V \mapsto \widehat{A}_2(V)$ is a bijection $R_j^F \rightarrow R_j^G$, and by Lemma 4(1) we have $q \in V \iff q' \in \widehat{A}_2(V)$. Therefore

$$|\{V \in R_j^F : q \in V\}| = |\{V' \in R_j^G : q' \in V'\}|.$$

Since this holds for all j , $\text{sig}_{\mathcal{R}_F}(q) = \text{sig}_{\mathcal{R}_G}(q')$. □

4.3 hyperplanes signatures

First, for any $U \in V_q(n)$ let

$$\text{sig}_{\mathcal{P}}(U) = (|U \cap P_1|, \dots, |U \cap P_k|).$$

For any $V \in V_q(m)$ let

$$\text{sig}_{\mathcal{Q}}(V) = (|V \cap Q_1|, \dots, |V \cap Q_{k'}|).$$

Lemma 6 (Validity of $\text{sig}_{\mathcal{P}}$ and $\text{sig}_{\mathcal{Q}}$)

- (i) Assume $(\mathcal{P}_F, \mathcal{P}_G)$ satisfies **Input point-cell invariant**. Then $(\text{sig}_{\mathcal{P}_F}, \text{sig}_{\mathcal{P}_G})$ is a valid domain-hyperplane signature pair.
- (ii) Assume $(\mathcal{Q}_F, \mathcal{Q}_G)$ satisfies **Output point-cell invariant**. Then $(\text{sig}_{\mathcal{Q}_F}, \text{sig}_{\mathcal{Q}_G})$ is a valid codomain-hyperplane signature pair.

Proof (i). Let (A_1, A_2) be a partial map, $(\widehat{A}_1, \widehat{A}_2)$ any extension, and suppose $\widehat{A}_1(U) = U'$. Fix i . Since $\widehat{A}_1$ is a bijection and respects the point cells, it maps P_i^F bijectively onto P_i^G . By Lemma 4(1), for $p \in \mathbb{F}_q^n$ we have $p \in U \iff \widehat{A}_1(p) \in U'$. Hence $\widehat{A}_1$ restricts to a bijection $U \cap P_i^F \rightarrow U' \cap P_i^G$, so $|U \cap P_i^F| = |U' \cap P_i^G|$ for all i , i.e., $\text{sig}_{\mathcal{P}_F}(U) = \text{sig}_{\mathcal{P}_G}(U')$.

(ii). Similarly, if $\widehat{A}_2(V) = V'$, then for each j the map $\widehat{A}_2$ sends Q_j^F bijectively onto Q_j^G , and Lemma 4(1) (in the codomain) gives $q \in V \iff \widehat{A}_2(q) \in V'$. Thus $|V \cap Q_j^F| = |V' \cap Q_j^G|$ for all j , i.e., $\text{sig}_{\mathcal{Q}_F}(V) = \text{sig}_{\mathcal{Q}_G}(V')$. $\square$

We now define signatures that transfer information between the domain and codomain via flats.

For any $U \in V_q(n)$ define

$$\text{sig}_{\mathcal{R}}(U) = (\{|U \cap F^{-1}(V)| : V \in R_1\}^*, \dots, \{|U \cap F^{-1}(V)| : V \in R_r\}^*).$$

Similarly, for any $V \in V_q(m)$ define

$$\text{sig}_{\mathcal{L}}(V) = (\{|F^{-1}(V) \cap U| : U \in L_1\}^*, \dots, \{|F^{-1}(V) \cap U| : U \in L_l\}^*).$$

Lemma 7 (Validity of $\text{sig}_{\mathcal{R}}$ and $\text{sig}_{\mathcal{L}}$)

- (i) Assume $(\mathcal{R}_F, \mathcal{R}_G)$ satisfies [Codomain hyperplane invariant](#). Then $(\text{sig}_{\mathcal{R}_F}, \text{sig}_{\mathcal{R}_G})$ is a valid domain-hyperplane signature pair.
- (ii) Assume $(\mathcal{L}_F, \mathcal{L}_G)$ satisfies [Domain hyperplane invariant](#). Then $(\text{sig}_{\mathcal{L}_F}, \text{sig}_{\mathcal{L}_G})$ is a valid codomain-hyperplane signature pair.

Proof (i). Let $(\widehat{A}_1, \widehat{A}_2)$ be any extension and suppose $\widehat{A}_1(U) = U'$. Fix j and write $V' := \widehat{A}_2(V)$ for $V \in R_j^F$. By codomain-hyperplane alignment, $V \mapsto V'$ is a bijection $R_j^F \rightarrow R_j^G$. Lemma 4(3) gives $|U \cap F^{-1}(V)| = |U' \cap G^{-1}(V')|$ for all such V , so the j -th multiset coordinate agrees. Hence $\text{sig}_{\mathcal{R}_F}(U) = \text{sig}_{\mathcal{R}_G}(U')$.

(ii). Let $\widehat{A}_2(V) = V'$. Fix i and write $U' := \widehat{A}_1(U)$ for $U \in L_i^F$. By domain-hyperplane alignment, $U \mapsto U'$ is a bijection $L_i^F \rightarrow L_i^G$. Lemma 4(3) gives $|F^{-1}(V) \cap U| = |G^{-1}(V') \cap U'|$ for all such U , so the i -th multiset coordinate agrees. Hence $\text{sig}_{\mathcal{L}_F}(V) = \text{sig}_{\mathcal{L}_G}(V')$. $\square$

We now have the 6 total signatures we need and will build on.

4.4 Compressed signatures

Having now established the validity of the signatures lets us now consider how to compute them. We are generally not going to compute the full signature as this is two computationally demanding. Consider the signatures $\text{sig}_{\mathcal{P}_F}$ and $\text{sig}_{\mathcal{P}_G}$ for the domain hyperplanes for example, we need to compute $\text{sig}_{\mathcal{P}_F}(U)$ and $\text{sig}_{\mathcal{P}_G}(U)$ for all $U \in \mathcal{V}_q(n)$, there are q^n such hyperplanes U and computing $\text{sig}_{\mathcal{P}_F}$ we need to consider all q^n point, so it total this is around $\mathcal{O}(q^{2n})$. We would like a better way.

We will do this by defining compressed version of our signature. The idea is that

Let $\varphi : \mathbb{N} \rightarrow \mathbb{Z}$ be any fixed hash function.

Then let $\pi_k : \mathbb{N}^k \rightarrow \mathbb{Z}$ be a weighted sum defined as

$$\pi_k(a_1, \dots, a_k) := \sum_{i=1}^k \varphi(i) \cdot a_i.$$

Using π_k we define scalar (compressed) versions of our vector-valued signatures by composition. Writing $k = |\mathcal{P}|$, $k' = |\mathcal{Q}|$, $r = |\mathcal{R}|$, and $\ell = |\mathcal{L}|$, set

$$\begin{aligned} \overline{sig}_{\mathcal{P}}(U) &:= \pi_k(sig_{\mathcal{P}}(U)), & \overline{sig}_{\mathcal{Q}}(V) &:= \pi_{k'}(sig_{\mathcal{Q}}(V)), \\ \overline{sig}_{\mathcal{L}}(p) &:= \pi_{\ell}(sig_{\mathcal{L}}(p)), & \overline{sig}_{\mathcal{R}}(q) &:= \pi_r(sig_{\mathcal{R}}(q)). \end{aligned}$$

Example. Suppose $|\mathcal{P}| = k = 3$ and for some domain hyperplane U we have $sig_{\mathcal{P}}(U) = (|U \cap P_1|, |U \cap P_2|, |U \cap P_3|) = (2, 5, 1)$. Then

$$\overline{sig}_{\mathcal{P}}(U) = \pi_3(2, 5, 1) = \varphi(1) \cdot 2 + \varphi(2) \cdot 5 + \varphi(3) \cdot 1.$$

Likewise, if $|\mathcal{L}| = \ell = 3$ and $sig_{\mathcal{L}}(p) = (0, 3, 1)$, then

$$\overline{sig}_{\mathcal{L}}(p) = \pi_3(0, 3, 1) = \varphi(2) \cdot 3 + \varphi(3) \cdot 1.$$

For the hyperplane signatures $sig_{\mathcal{R}}(U) = (M_1, \dots, M_r)$ and $sig_{\mathcal{L}}(V) = (N_1, \dots, N_{\ell})$, whose components are multisets, we additionally compress each multiset. For a finite multiset $M = \{m_1, \dots, m_t\} \subseteq \mathbb{N}$, define

$$\tau(M) := \sum_{i=1}^t m_i.$$

Applying τ component-wise and then π , define

$$\begin{aligned} \overline{sig}_{\mathcal{R}}(U) &:= \pi_r(\tau(M_1), \dots, \tau(M_r)), \\ \overline{sig}_{\mathcal{L}}(V) &:= \pi_{\ell}(\tau(N_1), \dots, \tau(N_{\ell})). \end{aligned}$$

The following is then follows from Lemma 2 and the definitions of the compressed signatures.

Lemma 8 (Validity of the compressed signatures) Each compressed signature pair is valid whenever the corresponding uncompressed signature pair is valid (i.e., if (sig_F, sig_G) is valid, then so is $(\overline{sig}_F, \overline{sig}_G)$). In particular, under the same hypotheses as the relevant validity lemmas in the previous subsections, the following compressed pairs are valid:

- $(\overline{sig}_{\mathcal{P}_F}, \overline{sig}_{\mathcal{P}_G})$ (domain hyperplanes via domain point partition $\mathcal{P}$),
- $(\overline{sig}_{\mathcal{Q}_F}, \overline{sig}_{\mathcal{Q}_G})$ (codomain hyperplanes via codomain point partition $\mathcal{Q}$),
- $(\overline{sig}_{\mathcal{L}_F}, \overline{sig}_{\mathcal{L}_G})$ (domain points),
- $(\overline{sig}_{\mathcal{R}_F}, \overline{sig}_{\mathcal{R}_G})$ (domain points),
- $(\overline{sig}_{\mathcal{R}_F}, \overline{sig}_{\mathcal{R}_G})$ (domain hyperplanes U),

- $(\overline{sig}_{\mathcal{L}_F}, \overline{sig}_{\mathcal{L}_G})$ (codomain hyperplanes V).

The point of introducing these compressed signatures is computational: they can be evaluated faster than the full signatures by passing to the Fourier domain.

We will utilize what is called the Radon transform. For a function $w : \mathbb{F}_q^n \rightarrow \mathbb{Z}$, its Radon transform $\mathcal{T}w$ on affine hyperplanes $H \in \mathcal{V}_q(n)$ is

$$\mathcal{T}w(H) = \sum_{x \in H} w(x).$$

We will also use its adjoint (backprojection): for a function g on hyperplanes,

$$\mathcal{T}^*g(x) = \sum_{H \ni x} g(H).$$

These will be applied to weight functions induced by ordered partitions. For the ordered partition $\mathcal{P} = (P_1, \dots, P_k)$ of the domain points, and the ordered partition $\mathcal{Q} = (Q_1, \dots, Q_{k'})$ of the codomain points, define

$$w_{\mathcal{P}}(x) = \varphi(i) \quad \text{if } x \in P_i, \quad w_{\mathcal{Q}}(y) = \varphi(j) \quad \text{if } y \in Q_j.$$

Likewise, for the ordered hyperplane partitions $\mathcal{L} = (L_1, \dots, L_\ell)$ and $\mathcal{R} = (R_1, \dots, R_r)$, define weight functions on hyperplanes by

$$w_{\mathcal{L}}(U) = \varphi(i) \quad \text{if } U \in L_i, \quad w_{\mathcal{R}}(V) = \varphi(j) \quad \text{if } V \in R_j.$$

Let $F : \mathbb{F}_q^n \rightarrow \mathbb{F}_q^m$. For a weight function $w : \mathbb{F}_q^n \rightarrow \mathbb{Z}$, define the pushforward along F by

$$(F_{\rightarrow}w)(y) = \sum_{x:F(x)=y} w(x),$$

and for a function $h : \mathbb{F}_q^m \rightarrow \mathbb{Z}$, define the pullback by

$$(F_{\leftarrow}h)(x) = h(F(x)).$$

Lemma 9 (Radon transform realizes the compressed signatures) With notation as above, each of the compressed signatures can be written as a (back)projection/Radon transform of an appropriate weight function. In particular,

$$\begin{aligned} \overline{sig}_{\mathcal{P}}(U) &= (\mathcal{T}w_{\mathcal{P}})(U), \\ \overline{sig}_{\mathcal{Q}}(V) &= (\mathcal{T}w_{\mathcal{Q}})(V), \\ \overline{sig}_{\mathcal{L}}(p) &= (\mathcal{T}^*w_{\mathcal{L}})(p), \\ \overline{sig}_{\mathcal{R}}(q) &= (\mathcal{T}^*w_{\mathcal{R}})(q), \\ \overline{sig}_{\mathcal{R}}(U) &= \mathcal{T}\left(F_{\leftarrow}(\mathcal{T}^*w_{\mathcal{R}})\right)(U), \end{aligned}$$

$$\overline{sig}_{\mathcal{L}}(V) = \mathcal{T}\left(F_{\rightarrow}(\mathcal{T}^*w_{\mathcal{L}})\right)(V).$$

Proof We prove each identity by expanding definitions.

(1) Domain hyperplanes via $\mathcal{P}$. For $U \in \mathcal{V}_q(n)$,

$$\overline{sig}_{\mathcal{P}}(U) = \pi_k(sig_{\mathcal{P}}(U)) = \sum_{i=1}^k i \cdot |U \cap P_i| = \sum_{x \in U} w_{\mathcal{P}}(x) = (\mathcal{T}w_{\mathcal{P}})(U).$$

(2) Codomain hyperplanes via $\mathcal{Q}$. For $V \in \mathcal{V}_q(m)$,

$$\overline{sig}_{\mathcal{Q}}(V) = \pi_{k'}(sig_{\mathcal{Q}}(V)) = \sum_{j=1}^{k'} j \cdot |V \cap Q_j| = \sum_{y \in V} w_{\mathcal{Q}}(y) = (\mathcal{T}w_{\mathcal{Q}})(V).$$

(3) Points via $\mathcal{L}$. For $p \in \mathbb{F}_q^n$, by definition of $w_{\mathcal{L}}$ on hyperplanes,

$$\overline{sig}_{\mathcal{L}}(p) = \pi_{\ell}(sig_{\mathcal{L}}(p)) = \sum_{i=1}^{\ell} i \cdot |\{U \in L_i : p \in U\}| = \sum_{U \ni p} w_{\mathcal{L}}(U) = (\mathcal{T}^*w_{\mathcal{L}})(p).$$

(4) Codomain points via $\mathcal{R}$. For $q \in \mathbb{F}_q^m$, we expand:

$$\overline{sig}_{\mathcal{R}}(q) = \pi_r(sig_{\mathcal{R}}(q)) = \sum_{V \ni q} w_{\mathcal{R}}(V) = (\mathcal{T}^*w_{\mathcal{R}})(q).$$

(5) Domain hyperplanes via $\mathcal{R}$. Let $U \in \mathcal{V}_q(n)$ and write $sig_{\mathcal{R}}(U) = (M_1, \dots, M_r)$ as in the definition. Then

$$\begin{aligned} \overline{sig}_{\mathcal{R}}(U) &= \pi_r(\tau(M_1), \dots, \tau(M_r)) = \sum_{j=1}^r j \cdot \sum_{V \in R_j} |U \cap F^{-1}(V)| \\ &= \sum_{V \in \mathcal{V}_q(m)} w_{\mathcal{R}}(V) |U \cap F^{-1}(V)| = \sum_{x \in U} \sum_{V \ni F(x)} w_{\mathcal{R}}(V). \end{aligned}$$

The inner sum is exactly $(\mathcal{T}^*w_{\mathcal{R}})(F(x)) = \overline{sig}_{\mathcal{R}}(F(x))$, hence the right-hand side equals

$$\sum_{x \in U} F_{\leftarrow}(\mathcal{T}^*w_{\mathcal{R}})(x) = \mathcal{T}\left(F_{\leftarrow}(\mathcal{T}^*w_{\mathcal{R}})\right)(U).$$

(6) Codomain hyperplanes via $\mathcal{L}$. Let $V \in \mathcal{V}_q(m)$ and write $sig_{\mathcal{L}}(V) = (N_1, \dots, N_{\ell})$. Then

$$\begin{aligned} \overline{sig}_{\mathcal{L}}(V) &= \pi_{\ell}(\tau(N_1), \dots, \tau(N_{\ell})) = \sum_{i=1}^{\ell} i \cdot \sum_{U \in L_i} |F^{-1}(V) \cap U| \\ &= \sum_{U \in \mathcal{V}_q(n)} w_{\mathcal{L}}(U) |F^{-1}(V) \cap U| = \sum_{x \in F^{-1}(V)} \sum_{U \ni x} w_{\mathcal{L}}(U). \end{aligned}$$

The inner sum is $(\mathcal{T}^*w_{\mathcal{L}})(x)$, so

$$\overline{sig}_{\mathcal{L}}(V) = \sum_{x \in F^{-1}(V)} (\mathcal{T}^*w_{\mathcal{L}})(x) = \sum_{y \in V} \sum_{x: F(x)=y} (\mathcal{T}^*w_{\mathcal{L}})(x) = \mathcal{T}\left(F_{\rightarrow}(\mathcal{T}^*w_{\mathcal{L}})\right)(V).$$

□

Now the question is naturally how do we compute these compressed signatures, we will not compute these function for one input at the time, but rather for all simultaneously. As an example lets consider how to compute $\overline{sig}_{\mathcal{R}}(U) = \mathcal{T}(F_{\leftarrow}(\mathcal{T}^*w_{\mathcal{R}}))$.

We then

1. We start with $w_{\mathcal{R}} : \mathcal{V}_q(m) \rightarrow \mathbb{Z}$
2. Apply $\mathcal{T}^*$ to get a function $\mathbb{F}_q^m \rightarrow \mathbb{Z}$
3. apply $F_{\leftarrow}$ to get a function $\mathbb{F}_q^n \rightarrow \mathbb{Z}$
4. Apply $\mathcal{T}$ to get a function $\mathcal{V}_q(n) \rightarrow \mathbb{Z}$

We can of course do step (2) in time $\mathcal{O}(q^n)$ the same is true when we need to apply $F_{\rightarrow}$. The only think we need here is to compute $\mathcal{T}$ and $\mathcal{T}^*$ efficiently.

Finally, we record the compressed signatures for each of the four objects we work with:

- **Domain points:** for $p \in \mathbb{F}_q^n$, we use the scalar invariant $\overline{sig}_{\mathcal{L}}(p) \in \mathbb{Z}$.
- **Codomain points:** for $q \in \mathbb{F}_q^m$, we use the scalar invariant $\overline{sig}_{\mathcal{R}}(q) \in \mathbb{Z}$.
- **Domain hyperplanes:** for $U \in \mathcal{V}_q(n)$, we use the pair

$$\overline{sig}_{\mathcal{P}, \mathcal{R}}(U) := (\overline{sig}_{\mathcal{P}}(U), \overline{sig}_{\mathcal{R}}(U)) \in \mathbb{Z}^2.$$

- **Codomain hyperplanes:** for $V \in \mathcal{V}_q(m)$, we use the pair

$$\overline{sig}_{\mathcal{Q}, \mathcal{L}}(V) := (\overline{sig}_{\mathcal{Q}}(V), \overline{sig}_{\mathcal{L}}(V)) \in \mathbb{Z}^2.$$

4.5 Computing the radon transform

Fix some additive character $\psi : \mathbb{F}_q \rightarrow \mathbb{C}$. For any function $f : \mathbb{F}_q^n \rightarrow \mathbb{C}$ the additive Fourier transform is defined as

$$\widehat{f}(\alpha) = \sum_{x \in \mathbb{F}_q^n} f(x) \psi(-\alpha \cdot x)$$

If we let X contain one representative for each 1 dimensional subspace in $\mathbb{F}_q^n$ then each affine hyperplane in $\mathbb{F}_q^n$ can be written as $H_{a,b} = \{x \in \mathbb{F}_q^n : a \cdot x = b\}$, the key identity that we use then is the slice formula

$$\mathcal{T}f(H_{a,b}) = \frac{1}{q} \sum_{t \in \mathbb{F}_q} \widehat{f}(ta) \psi(tb)$$

Using this to compute $\mathcal{T}f$ we just need to the following:

Using the Fourier slice formula, $\mathcal{T}f$ can be computed as follows.

1. Compute the d -dimensional Fourier transform

$$\widehat{f}(\alpha) = \sum_{x \in \mathbb{F}_q^d} f(x) \psi(-\alpha \cdot x), \quad \alpha \in \mathbb{F}_q^d.$$

2. For each $a \in \Sigma_d$, form the sequence

$$t \mapsto \widehat{f}(ta), \quad t \in \mathbb{F}_q.$$

3. For each $a \in \Sigma_d$, apply the inverse 1-dimensional Fourier transform in the t -variable to obtain

$$(\mathcal{T}f)(H_{a,b}) = \frac{1}{q} \sum_{t \in \mathbb{F}_q} \widehat{f}(ta) \psi(tb), \quad b \in \mathbb{F}_q.$$

4. Since every affine hyperplane is uniquely of the form $H_{a,b}$ with $a \in \Sigma_d$ and $b \in \mathbb{F}_q$, this determines $\mathcal{T}f$ on all affine hyperplanes.
for the adjoint, for any function $g : V_q(n) \rightarrow \mathbb{Z}$ let the 1 dim Fourier transform be

$$\widehat{g}_a(t) = \sum_{b \in \mathbb{F}_q} g(H_{a,b}) \psi(-tb)$$

then

$$\widehat{T^*g}(ta) = q^{d-1} \widehat{g}_a(t)$$

So we get $\mathcal{T}g$ by the inverse Fourier transform.

To compute $\mathcal{T}^*g$, we first compute its Fourier transform and then invert.

1. For each $a \in \Sigma_d$, form the sequence

$$b \mapsto g(H_{a,b}), \quad b \in \mathbb{F}_q.$$

2. For each $a \in \Sigma_d$, compute its 1-dimensional Fourier transform

$$\widehat{g}_a(t) = \sum_{b \in \mathbb{F}_q} g(H_{a,b}) \psi(-tb), \quad t \in \mathbb{F}_q.$$

3. Define the Fourier transform of $h = \mathcal{T}^*g$ by

$$\widehat{h}(ta) = q^{d-1} \widehat{g}_a(t) \quad \text{for } a \in \Sigma_d, t \in \mathbb{F}_q^\times,$$

and

$$\widehat{h}(0) = q^{d-1} \sum_H g(H).$$

4. Apply the inverse d -dimensional Fourier transform to $\widehat{h}$. The result is

$$h = \mathcal{T}^*g.$$

We can now efficiently compute T and T^* and therefore all the compressed signatures efficiently, i.e in time $\mathcal{O}(q^n)$ or $\mathcal{O}(q^m)$.

5 Partition refinement

The first thing is given a pair of partitions and signatures, split the cells based on the signature and return false if the corresponding cell in each partition don't have the same size.

Algorithm 1: $\text{REFINE}((\mathcal{P}_F = (B_1^F, \dots, B_k^F), \mathcal{P}_G = (B_1^G, \dots, B_k^G)), (\text{sig}_F, \text{sig}_G))$

Output: Either **false**, or **true** and ordered partitions $(\mathcal{P}'_F, \mathcal{P}'_G)$ with the same shape.

```

 $\mathcal{P}'_F, \mathcal{P}'_G \leftarrow ()$ ;
for  $i \leftarrow 1$  to  $k$  do
    // Group elements of each block by their signature values
    Build buckets  $\mathcal{B}_F[s] := \{x \in B_i^F : \text{sig}_F(x) = s\}$  for each signature value  $s$ 
    occurring on  $B_i^F$ ;
    Build buckets  $\mathcal{B}_G[s] := \{y \in B_i^G : \text{sig}_G(y) = s\}$  for each signature value  $s$ 
    occurring on  $B_i^G$ ;
    // Fail if the refined blocks would no longer match in shape
    if  $\{(|\mathcal{B}_F[s]|, s) : \mathcal{B}_F[s] \neq \emptyset\} \neq \{(|\mathcal{B}_G[s]|, s) : \mathcal{B}_G[s] \neq \emptyset\}$  then
        return false;
    // Append refined sub-blocks (no sorting required)
    Let  $S$  be the list of signature values occurring in  $B_i^F$ , in order of first
    occurrence;
    foreach  $s \in S$  do
        Append  $\mathcal{B}_F[s]$  to  $\mathcal{P}'_F$  and append  $\mathcal{B}_G[s]$  to  $\mathcal{P}'_G$ ;
return  $(\mathcal{P}'_F, \mathcal{P}'_G)$ ;

```

5.1 Partition refinement

Now are simply going to keep refining until it is no longer possible.

Algorithm 2: PARTITIONREFINEMENT($(\mathcal{P}_F, \mathcal{P}_G), (\mathcal{Q}_F, \mathcal{Q}_G), (\mathcal{L}_F, \mathcal{L}_G), (\mathcal{R}_F, \mathcal{R}_G)$)

Output: Either *false*, or refined partitions with matching shapes.

```

Changed  $\leftarrow$  true;
while Changed do
    Changed  $\leftarrow$  false;
     $(\mathcal{P}'_F, \mathcal{P}'_G) \leftarrow \text{REFINE}((\mathcal{P}_F, \mathcal{P}_G), \overline{\text{sig}}_{\mathcal{L}_F, \mathcal{R}_F}, \overline{\text{sig}}_{\mathcal{L}_G, \mathcal{R}_G});$ 
    if  $(\mathcal{P}'_F, \mathcal{P}'_G) = \text{false}$  then
        return false;
    if  $(\mathcal{P}'_F, \mathcal{P}'_G) \neq (\mathcal{P}_F, \mathcal{P}_G)$  then
        Changed  $\leftarrow$  true;
     $(\mathcal{P}_F, \mathcal{P}_G) \leftarrow (\mathcal{P}'_F, \mathcal{P}'_G);$ 
     $(\mathcal{Q}'_F, \mathcal{Q}'_G) \leftarrow \text{REFINE}((\mathcal{Q}_F, \mathcal{Q}_G), \overline{\text{sig}}_{\mathcal{L}_F, \mathcal{R}_F}, \overline{\text{sig}}_{\mathcal{L}_G, \mathcal{R}_G});$ 
    if  $(\mathcal{Q}'_F, \mathcal{Q}'_G) = \text{false}$  then
        return false;
    if  $(\mathcal{Q}'_F, \mathcal{Q}'_G) \neq (\mathcal{Q}_F, \mathcal{Q}_G)$  then
        Changed  $\leftarrow$  true;
     $(\mathcal{Q}_F, \mathcal{Q}_G) \leftarrow (\mathcal{Q}'_F, \mathcal{Q}'_G);$ 
     $(\mathcal{L}'_F, \mathcal{L}'_G) \leftarrow \text{REFINE}((\mathcal{L}_F, \mathcal{L}_G), \overline{\text{sig}}_{\mathcal{P}_F, \mathcal{R}_F}, \overline{\text{sig}}_{\mathcal{P}_G, \mathcal{R}_G});$ 
    if  $(\mathcal{L}'_F, \mathcal{L}'_G) = \text{false}$  then
        return false;
    if  $(\mathcal{L}'_F, \mathcal{L}'_G) \neq (\mathcal{L}_F, \mathcal{L}_G)$  then
        Changed  $\leftarrow$  true;
     $(\mathcal{L}_F, \mathcal{L}_G) \leftarrow (\mathcal{L}'_F, \mathcal{L}'_G);$ 
     $(\mathcal{R}'_F, \mathcal{R}'_G) \leftarrow \text{REFINE}((\mathcal{R}_F, \mathcal{R}_G), \overline{\text{sig}}_{\mathcal{P}_F, \mathcal{L}_F}, \overline{\text{sig}}_{\mathcal{P}_G, \mathcal{L}_G});$ 
    if  $(\mathcal{R}'_F, \mathcal{R}'_G) = \text{false}$  then
        return false;
    if  $(\mathcal{R}'_F, \mathcal{R}'_G) \neq (\mathcal{R}_F, \mathcal{R}_G)$  then
        Changed  $\leftarrow$  true;
     $(\mathcal{R}_F, \mathcal{R}_G) \leftarrow (\mathcal{R}'_F, \mathcal{R}'_G);$ 
return true,  $(\mathcal{P}_F, \mathcal{P}_G), (\mathcal{Q}_F, \mathcal{Q}_G), (\mathcal{L}_F, \mathcal{L}_G), (\mathcal{R}_F, \mathcal{R}_G);$ 

```

6 Using automorphism groups

Our algorithms will take advantage of the automorphisms to avoid branching. This is based on the following Lemma.

Lemma 10 (Orbit pruning via automorphisms on points) Let $F, G : \mathbb{F}_q^n \rightarrow \mathbb{F}_q^m$, let (A_1, A_2) be a partial map, and let $H \leq \text{Aut}(G)$.

- (i) Fix $x \in \mathbb{F}_q^n \setminus \text{dom}(A_1)$. If $y, y' \in \mathbb{F}_q^n$ lie in the same orbit under $\text{Stab}_{A_1}(H)$ and there exists an extension $(\widetilde{A_1}, \widetilde{A_2})$ of (A_1, A_2) with $\widetilde{A_1}(x) = y$, then there exists an extension $(\widetilde{A_1}, \widetilde{A_2})$ of (A_1, A_2) with $\widetilde{A_1}(x) = y'$.
- (ii) Fix $z \in \mathbb{F}_q^m \setminus \text{dom}(A_2)$. If $w, w' \in \mathbb{F}_q^m$ lie in the same orbit under $\text{Stab}_{A_2}(H)$ and there exists an extension $(\widetilde{A_1}, \widetilde{A_2})$ of (A_1, A_2) with $\widetilde{A_2}(z) = w$, then there exists an extension $(\widetilde{A_1}, \widetilde{A_2})$ of (A_1, A_2) with $\widetilde{A_2}(z) = w'$.

Proof (i) Choose $(B_1, B_2) \in \text{Stab}_{A_1}(H)$ such that $B_1(y) = y'$. Define $\widetilde{A_1} := B_1 \circ \widehat{A_1}$ and $\widetilde{A_2} := B_2 \circ \widehat{A_2}$. Then $(\widetilde{A_1}, \widetilde{A_2})$ is still an extension of (A_1, A_2) , and $\widetilde{A_1}(x) = B_1(\widehat{A_1}(x)) = y'$.
(ii) Choose $(B_1, B_2) \in \text{Stab}_{A_2}(H)$ such that $B_2(w) = w'$. Define $\widetilde{A_1} := B_1 \circ \widehat{A_1}$ and $\widetilde{A_2} := B_2 \circ \widehat{A_2}$. Then $(\widetilde{A_1}, \widetilde{A_2})$ is still an extension of (A_1, A_2) , and $\widetilde{A_2}(z) = B_2(\widehat{A_2}(z)) = w'$. $\square$

We have the same results when we branch on hyperplanes

Lemma 11 (Orbit pruning via automorphisms on hyperplanes) Let $F, G : \mathbb{F}_q^n \rightarrow \mathbb{F}_q^m$, let (A_1, A_2) be a partial map, and let $H \leq \text{Aut}(G)$.

- (i) Fix $U \in V_q(n)$. If $V, V' \in \mathbb{F}_q^n$ lie in the same orbit under $\text{Stab}_{A_1}(H)$ and there exists an extension $(\widetilde{A_1}, \widetilde{A_2})$ of (A_1, A_2) with $\widetilde{A_1}(U) = V$, then there exists an extension $(\widetilde{A_1}, \widetilde{A_2})$ of (A_1, A_2) with $\widetilde{A_1}(U) = V'$.
- (ii) Fix $U \in V_q(m)$. If $V, V' \in V_q(m)$ lie in the same orbit under $\text{Stab}_{A_2}(H)$ and there exists an extension $(\widetilde{A_1}, \widetilde{A_2})$ of (A_1, A_2) with $\widetilde{A_2}(U) = V$, then there exists an extension $(\widetilde{A_1}, \widetilde{A_2})$ of (A_1, A_2) with $\widetilde{A_2}(U) = V'$.

7 Affine automorphism algorithm

There are now 4 potential objects we branch on, the basic logic is the same for all of them.

Algorithm 3: $\text{BRANCH}(x, B, H_{\text{prev}}, H_{\text{new}})$

$\mathcal{O} = (O_1, \dots, O_s) = \text{ORBITPARTITION}(P_i^F,)$

We are now going to present the automorphism algorithm for linear and affine equivalence. The first is the depth first search part of the algorithm

Algorithm 4: DFS-AUTO $\left(A = (A_1, A_2), (\mathcal{P}_F, \mathcal{P}_G), (\mathcal{Q}_F, \mathcal{Q}_G), (\mathcal{L}_F, \mathcal{L}_G), (\mathcal{R}_F, \mathcal{R}_G), H_{\text{stab}}, H_{\text{glob}}\right)$

Input: Partial affine map (A_1, A_2) , ordered partitions $(\mathcal{P}_F, \mathcal{P}_G), (\mathcal{Q}_F, \mathcal{Q}_G), (\mathcal{L}_F, \mathcal{L}_G), (\mathcal{R}_F, \mathcal{R}_G)$, a current stabilizer subgroup H_{stab} for orbit pruning, and a global accumulator $H_{\text{glob}} \leq \text{Aut}(F)$.

Output: The updated global accumulator H_{glob} .

$(\text{IsConsistent}, ((\mathcal{P}_F, \mathcal{P}_G), (\mathcal{Q}_F, \mathcal{Q}_G), (\mathcal{L}_F, \mathcal{L}_G), (\mathcal{R}_F, \mathcal{R}_G))) \leftarrow$
 PARTITIONREFINEMENT $((\mathcal{P}_F, \mathcal{P}_G), (\mathcal{Q}_F, \mathcal{Q}_G), (\mathcal{L}_F, \mathcal{L}_G), (\mathcal{R}_F, \mathcal{R}_G));$

if not IsConsistent then

└ **return** H_{glob} ;

for $P_i^F \in I(\mathcal{P}_F)$ **do**

└ $x, y \leftarrow$ element in P_i^F and P_i^G respectively;

└ $\text{valid}, A_1 \leftarrow \text{UPDATE}(x, y, A_1);$

└ **if not valid then**

└└ **return** H_{glob}

for $Q_i^F \in I(\mathcal{Q}_F)$ **do**

└ $x, y \leftarrow$ element in Q_i^F and Q_i^G respectively;

└ $\text{valid}, A_2 \leftarrow \text{UPDATE}(x, y, A_2);$

└ **if not valid then**

└└ **return** H_{glob}

if A_1 and A_2 are fully determined **then**

└ **if** $G = A_2 \circ F \circ A_1$ **then**

└└ $H_{\text{glob}} \leftarrow \langle H_{\text{glob}}, (A_1, A_2) \rangle;$

└ **return** H_{glob} ;

Let P_i^F be the smallest cell in $\mathcal{P}_F$ where $|P_i^F \setminus \text{Dom}(A_1)| \geq 1$;

Let Q_j^F be the smallest cell in $\mathcal{Q}_F$ where $|Q_j^F \setminus \text{Dom}(A_2)| \geq 1$;

Let L_k^F be the smallest non-singleton cell in $\mathcal{L}_F$;

Let R_s^F be the smallest non-singleton cell in $\mathcal{R}_F$;

(continued on next page);

```

if  $|P_i^F| < |Q_i^F|$  then
   $\mathcal{O} = (O_1, \dots, O_s) = \text{ORBITPARTITION}(P_i^F, \pi_1(H_{\text{stab}}));$ 
   $B \leftarrow \emptyset;$ 
   $Y \leftarrow$  one element for each cell in  $\mathcal{O}$ .
  while  $Y \neq \emptyset$  do
    Choose any  $y \in Y$ ;
     $B \leftarrow B \cup \{y\};$ 
     $(\text{IsConsistent}, A'_1) \leftarrow \text{UPDATE}(A_1, x, y);$ 
    if IsConsistent then
      Individualize  $x$  and  $y$  in  $\mathcal{P}_F$  and  $\mathcal{P}_G$  to get  $\mathcal{P}'_F$  and  $\mathcal{P}'_G$ ;
       $H_{\text{glob}} \leftarrow \text{DFS-AUTO}\left(\begin{array}{l} (A'_1, A_2), (\mathcal{P}_F, \mathcal{P}_G), (\mathcal{Q}_F, \mathcal{Q}_G), \\ (\mathcal{L}_F, \mathcal{L}_G), (\mathcal{R}_F, \mathcal{R}_G), \text{Stab}_x(H_{\text{stab}}), H_{\text{glob}} \end{array}\right);$ 
      if  $\text{Stab}_{(A'_1, A_2)}(H_{\text{glob}}) > H_{\text{stab}}$  then
         $H_{\text{stab}} \leftarrow \text{Stab}_{(A'_1, A_2)}(H_{\text{glob}});$ 
         $\mathcal{O} = (O_1, \dots, O_s) \leftarrow \text{ORBITPARTITION}(P_i^G, \pi_1(H_{\text{stab}}));$ 
         $Y \leftarrow$  one element from each orbit  $O_j \in \mathcal{O}$  with  $O_j \cap B = \emptyset$ ;
    else
       $\mathcal{O} = (O_1, \dots, O_s) = \text{ORBITPARTITION}(Q_j^F, \pi_2(H_{\text{stab}}));$ 
       $B \leftarrow \emptyset;$ 
       $Y \leftarrow$  one element for each cell in  $\mathcal{O}$ .
      while  $Y \neq \emptyset$  do
        Choose any  $y \in Y$ ;
         $B \leftarrow B \cup \{y\};$ 
         $(\text{IsConsistent}, A'_2) \leftarrow \text{UPDATE}(A_2, x, y);$ 
        if IsConsistent then
          Individualize  $x$  and  $y$  in  $\mathcal{Q}_F$  and  $\mathcal{Q}_G$  to get  $\mathcal{Q}'_F$  and  $\mathcal{Q}'_G$ ;
           $H_{\text{glob}} \leftarrow \text{DFS-AUTO}\left(\begin{array}{l} (A_1, A'_2), (\mathcal{P}_F, \mathcal{P}_G), (\mathcal{Q}_F, \mathcal{Q}_G), \\ (\mathcal{L}_F, \mathcal{L}_G), (\mathcal{R}_F, \mathcal{R}_G), \text{Stab}_x(H_{\text{stab}}), H_{\text{glob}} \end{array}\right);$ 
          if  $\text{Stab}_{(A_1, A'_2)}(H_{\text{glob}}) > H_{\text{stab}}$  then
             $H_{\text{stab}} \leftarrow \text{Stab}_{(A_1, A'_2)}(H_{\text{glob}});$ 
             $\mathcal{O} = (O_1, \dots, O_s) \leftarrow \text{ORBITPARTITION}(Q_j^G, \pi_2(H_{\text{stab}}));$ 
             $Y \leftarrow$  one element from each orbit  $O_j \in \mathcal{O}$  with  $O_j \cap B = \emptyset$ ;
      return  $H_{\text{global}}$ 

```

Now the automorphism algorithm is simply.

Algorithm 4: AFFINE-AUTOMORPHISM(F)

Input: A polynomial map $F : \mathbb{F}_q^n \rightarrow \mathbb{F}_q^m$.

Output: A generating set for $\text{Aut}(F)$.

$A_1, A_2 \leftarrow$ empty partial maps;

$(\mathcal{P}_F, \mathcal{P}_G) \leftarrow ((\mathbb{F}_q^n), (\mathbb{F}_q^n))$; // initial domain partitions

$(\mathcal{Q}_F, \mathcal{Q}_G) \leftarrow ((\mathbb{F}_q^m), (\mathbb{F}_q^m))$; // initial codomain partitions

$(\mathcal{L}_F, \mathcal{L}_G) \leftarrow (V_q(n), V_q(n))$; // hyperplanes in $\mathbb{F}_q^n$

$(\mathcal{R}_F, \mathcal{R}_G) \leftarrow (V_q(m), V_q(m))$; // hyperplanes in $\mathbb{F}_q^m$

$H_{\text{glob}} \leftarrow \langle (\text{id}_{\mathbb{F}_q^n}, \text{id}_{\mathbb{F}_q^m}) \rangle$ $H_{\text{stab}} \leftarrow H_{\text{glob}}$

return DFS-AUTO $\left((A_1, A_2), (\mathcal{P}_F, \mathcal{P}_G), (\mathcal{Q}_F, \mathcal{Q}_G), \right.$
 $\left. (\mathcal{L}_F, \mathcal{L}_G), (\mathcal{R}_F, \mathcal{R}_G), H_{\text{stab}}, H_{\text{glob}} \right)$;

8 Affine equivalence algorithm

Algorithm 5: DFS-EQUIVALENCE $\left(\begin{smallmatrix} A = (A_1, A_2), (\mathcal{P}_F, \mathcal{P}_G), (\mathcal{Q}_F, \mathcal{Q}_G), \\ (\mathcal{L}_F, \mathcal{L}_G), (\mathcal{R}_F, \mathcal{R}_G), H_{\text{stab}}, H_{\text{glob}} \end{smallmatrix}\right)$

```

(IsConsistent, (( $\mathcal{P}_F, \mathcal{P}_G$ ), ( $\mathcal{Q}_F, \mathcal{Q}_G$ ), ( $\mathcal{L}_F, \mathcal{L}_G$ ), ( $\mathcal{R}_F, \mathcal{R}_G$ )))  $\leftarrow$ 
PARTITIONREFINEMENT(( $\mathcal{P}_F, \mathcal{P}_G$ ), ( $\mathcal{Q}_F, \mathcal{Q}_G$ ), ( $\mathcal{L}_F, \mathcal{L}_G$ ), ( $\mathcal{R}_F, \mathcal{R}_G$ ));
if not IsConsistent then
   $\mid$  return  $H_{\text{glob}}$ ;
for  $P_i^F \in I(\mathcal{P}_F)$  do
   $x, y \leftarrow$  element in  $P_i^F$  and  $P_i^G$  respectively;
  valid,  $A_1 \leftarrow$  UPDATE( $x, y, A_1$ );
  if not valid then
     $\mid$  return  $H_{\text{glob}}$ 
for  $Q_i^F \in I(\mathcal{Q}_F)$  do
   $x, y \leftarrow$  element in  $Q_i^F$  and  $Q_i^G$  respectively;
  valid,  $A_2 \leftarrow$  UPDATE( $x, y, A_2$ );
  if not valid then
     $\mid$  return  $H_{\text{glob}}$ 
if  $A_1$  and  $A_2$  are fully determined then
  if  $G = A_2 \circ F \circ A_1$  then
     $\mid$   $H_{\text{glob}} \leftarrow \langle H_{\text{glob}}, (A_1, A_2) \rangle$ ;
  return  $H_{\text{glob}}$ ;
Let  $P_i^F$  be the smallest cell in  $\mathcal{P}_F$  where  $|P_i^F \setminus \text{Dom}(A_1)| \geq 1$ ;
Let  $Q_j^F$  be the smallest cell in  $\mathcal{Q}_F$  where  $|Q_j^F \setminus \text{Dom}(A_2)| \geq 1$ ;
if  $|P_i^F| < |Q_j^F|$  then
   $\mathcal{O} = (O_1, \dots, O_s) \leftarrow$  ORBITPARTITION( $P_i^F \setminus \text{Dom}(A_1), \pi_1(H_{\text{stab}})$ );
  for  $O_i \in \mathcal{O}$  do
     $y \leftarrow$  any element of  $O_i$ ;
    Individualize  $x$  and  $y$  in  $\mathcal{Q}_F$  and  $\mathcal{Q}_G$  to get  $\mathcal{Q}'_F$  and  $\mathcal{Q}'_G$ ;
     $H_{\text{glob}} \leftarrow$  DFS-AUTO $\left(\begin{smallmatrix} (A_1, A'_2), (\mathcal{P}_F, \mathcal{P}_G), (\mathcal{Q}'_F, \mathcal{Q}'_G), \\ (\mathcal{L}_F, \mathcal{L}_G), (\mathcal{R}_F, \mathcal{R}_G), \text{Stab}_x(H_{\text{stab}}), H_{\text{glob}} \end{smallmatrix}\right)$ ;

```

References